

The Role of Volume Regulation and Thermoregulation in AKI during Marathon Running

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Abstract

Background and objectives Marathon runners develop transient AKI with urine sediments and injury biomarkers suggesting nephron damage.

Design, setting, participants, & measurements To investigate the etiology, we examined volume and thermoregulatory responses as possible mechanisms in runners' AKI using a prospective cohort of runners in the 2017 Hartford Marathon. Vitals, blood, and urine samples were collected in 23 runners 1 day premarathon and immediately and 1 day postmarathon. We measured copeptin at each time point. Continuous core body temperature, sweat sodium, and volume were assessed during the race. The primary outcome of interest was AKI, defined by AKIN criteria.

Results Runners ranged from 22 to 63 years old; 43% were men. Runners lost a median (range) of 2.34 (0.50–7.21) g of sodium and 2.47 (0.36–6.81) L of volume *via* sweat. After accounting for intake, they had a net negative sodium and volume balance at the end of the race. The majority of runners had increases in core body temperature to 38.4 (35.8–41)°C during the race from their baseline. Fifty-five percent of runners developed AKI, yet 74% had positive urine microscopy for acute tubular injury. Runners with more running experience and increased participation in prior marathons developed a rise in creatinine as compared with those with lesser experience. Sweat sodium losses were higher in runners with AKI versus non-AKI (median, 3.41 [interquartile range (IQR), 1.7–4.8] versus median, 1.4 [IQR, 0.97–2.8] g; $P=0.06$, respectively). Sweat volume losses were higher in runners with AKI versus non-AKI (median, 3.89 [IQR, 1.49–5.09] versus median, 1.66 [IQR, 0.72–2.84] L; $P=0.03$, respectively). Copeptin was significantly higher in runners with AKI versus those without (median, 79.9 [IQR, 25.2–104.4] versus median, 11.3 [IQR, 6.6–43.7]; $P=0.02$, respectively). Estimated temperature was not significantly different.

Conclusions All runners experienced a substantial rise in copeptin and body temperature along with salt and water loss due to sweating. Sodium and volume loss *via* sweat as well as plasma copeptin concentrations were associated with AKI in runners.

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Introduction

Half a million runners participate in marathons yearly (1); however, the association between marathon running and kidney function has largely been overlooked, because runners are regarded as healthy athletes with trained physiology to tolerate high states of energy expenditure (2,3). We previously investigated the effects of marathon running on kidney function and found that the majority of runners developed AKI (4). Unexpectedly, the elevations in injury biomarkers and urine microscopy abnormalities were similar in severity to those of patients in the intensive care unit (5). However, all runners had AKI resolution indicating transient kidney injury. The mechanisms leading to this severe but transient AKI remain unclear.

Because marathon running induces sustained physical stress and significant increases in metabolic rate, there is a substantial challenge to the regulatory responses

in the body. We measured copeptin (vasopressin prohormone) and estimated core body temperature and sweat sodium to assess the volume and thermoregulatory responses in marathon running. Copeptin concentrations reflect water regulation, and tonicity. Studies have shown that there is a significant rise in vasopressin with marathon running (6), but none have measured copeptin concentrations or have shown the association of vasopressin or copeptin with kidney injury in runners. Furthermore, given the kidney hemodynamics during a marathon, with blood shunting from the heart and kidneys to the skin and muscles (7) and vasopressin-stimulated vasoconstriction as well as dehydration and volume loss, we hypothesize that copeptin concentrations will be altered with marathon running and will be associated with AKI (8).

The thermoregulatory balance during marathon running reflects the rate of endogenous metabolic heat

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production and the exchange of heat with the environment (9,10). Thermoregulation during running involves vasodilation leading to heat exchange *via* blood from the core to the skin and sweating for evaporative heat loss. It is estimated that the amount of energy required to raise 1 kg of tissue by 1°C is 3.5 kJ/1°C per kg; therefore, for an average 70-kg runner, 245 kJ are needed for a 1°C increase (9). Without a heat loss mechanism in place, this energy buildup exposes the body to dangerous temperatures. These mechanisms rely on a large skin-to-ambient temperature gradient, allowing heat dissipation from vasodilated vessels (11). As core body temperature rises, vasodilation for heat dissipation also increases, and therefore, we measured continuous core body temperature throughout the race as an indirect measurement of skin vasodilation and the primary response to thermoregulation. To prevent dangerous rises in core body temperature, heat loss is also facilitated by sweat evaporation during running (12). Core body temperature and sweat volume and sodium reflect increased metabolic and physical demands on the runners, which in turn, lead to shunting of blood away from main organs to increase blood flow to the skin for heat dissipation. To further examine the mechanism behind AKI in runners, we tested multiple variables related to volume and thermoregulation in this prospective observational study evaluating 23 runners who participated in the Hartford Marathon in Connecticut in 2017. Understanding the mechanisms behind runners' AKI may aid in decreasing the incidence of AKI and possibly prevent long-term maladaptive changes in the kidneys.

Materials and Methods

Study Design and Participants

Twenty-three marathon runners participating in the 2017 Hartford Marathon (Connecticut) were enrolled in this study (only one participant overlapped from our 2015 study) (4). One runner could not provide blood samples and was excluded from analysis. Runners were recruited *via* flyers posted at local running clubs and announcements on the Hartford Marathon Registration website and social media, including Twitter and Facebook. Inclusion criteria included experienced runners >21 years old with body mass index of 18.5–24.9, with at least 2 years of running experience, with a

minimum of 15 miles of training per week for the last 3 years, and who completed at least four races >20 km in distance. Runners were excluded from the study if they sustained any major musculoskeletal running-related injuries over the last 4 months, participated in another marathon within 4 weeks before race, used nonsteroidal anti-inflammatory drugs within 48 hours before or 24 hours after the marathon, used statins, or used anabolic steroids.

Primary Variables

Sample Collection and Measurement. Urine and blood samples were collected at three time points: 24 hours premarathon (day 0), immediately (within 30 minutes) postmarathon (day 1), and 24 hours postmarathon (day 2) as shown in Figure 1. Serum basic metabolic panels, including sodium and creatinine, creatine phosphokinase (CPK), hemoglobin, copeptin, urine protein, and urine microscopy, were evaluated at each time point. Sweat patches from PharmChek (13) were used to collect sweat at the 5-mile mark during the race. A patch was placed on a runner's forearm before the race and removed at the 5-mile mark. We decided to remove patches at this time point on the basis of our pilot studies, because the patches became saturated with sweat further into the race. We estimated total sweat volume (liters) and sodium (grams) using the equations displayed in Supplemental Table 1.

Collected samples were transported on ice to the Yale University biorepository, where they were centrifuged at 5000 rpm for 10 minutes at 4°C, separated into 1-ml aliquots, and immediately stored at –80°C until measurement. All laboratory personnel were blinded to participant information.

Vital Signs and Clinical Laboratory Measurements. BP, heart rate, and weight were measured on days 0, 1, and 2. The Zephyr (14,15) bioharness device was used on day 1 during the entire race to collect continuous second by second body temperature measurements. The Zephyr calculated body temperature on the basis of a heart rate algorithm (16). Previous studies have shown a correlation between core body temperature measurements using a heart rate algorithm and ingestible thermometer pills (17). This device was strapped around the upper abdomen/chest of each runner.

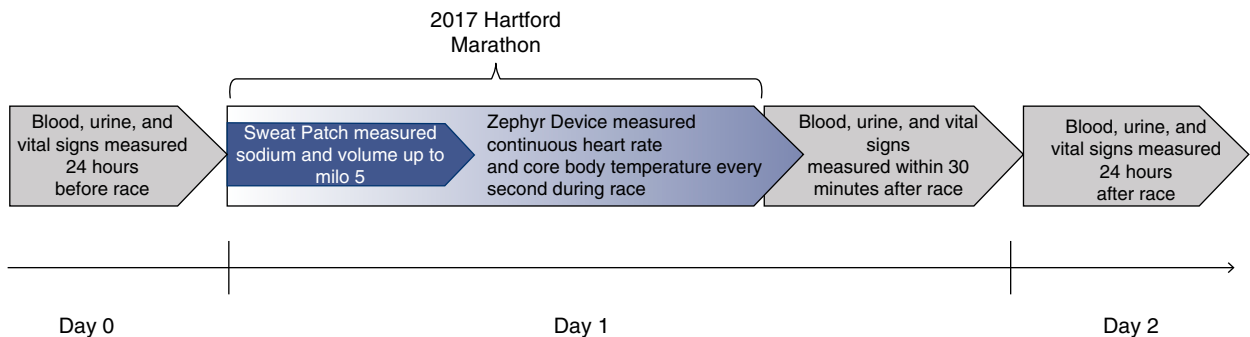


Figure 1. | Sample collection occurred 24 hours before the race, during and 30 minutes after the race, and 24 hours after the race. Blood, urine, and vital signs were measured at 1 day before the race, immediately after the race, and 1 day after the race. Sweat was only collected at the 5-mile mark during the race, and the Zephyr device measured continuous heart rate throughout the race and calculated core body temperature every second during the marathon.

B·R·A·H·M·S Copeptin proAVP KRYPTOR was measured on KRYPTOR Compact PLUS analyzers using the commercial sandwich immunofluorescence assay (B·R·A·H·M·S GmbH, Hennigsdorf, Germany). Coefficients of variance for copeptin are shown in Supplemental Table 2. Yale–New Haven Hospital's Clinical Chemistry Laboratory measured urine sodium, urine creatinine, and sweat electrolytes on the Roche Modular Analytics. Urine test strips/dipsticks were used for urinalysis *via* an automated analyzer by Siemens Clinitek diagnostics. Basic metabolic profile, serum creatinine, and serum CPK were measured, and urine microscopy was performed as described in our prior publication (4). Participants were given paper surveys with detailed questions regarding comorbidities, training habits premarathon, and details regarding oral intake the day of the marathon. All calculations of fluid and salt intake were estimated using manufacturer labels.

Secondary Variables: Novel Biomarkers of Kidney Injury

The following biomarkers were measured to validate our prior findings (4) in the urine: IL-18, kidney injury molecule-1, neutrophil gelatinase-associated lipocalin (NGAL), monocyte chemoattractant protein-1 (MCP-1), chitinase-3-like protein 1 (YKL-40). The following biomarkers were measured to validate our prior findings in the blood: IFN, IL-1 β , IL-2, IL-4, IL-6, IL-8, IL-10, IL-12, IL-13, IL-18, kidney injury molecule-1, NGAL, TNF- α , MCP-1, and YKL-40. Urinary and serum biomarker measurements were analyzed as concentrations of nanograms per milliliter or picograms per milliliter. All were measured using the Meso Scale Discovery platform (MSD, Gaithersburg, MD). We have previously used MSD technology to measure several kidney injury biomarkers (4). Values below the limit of detection were excluded from analysis. Coefficients of variance for novel biomarkers are shown in Supplemental Table 2.

Outcome Definitions

AKI was defined as stage 1 or higher using the AKI Network criteria (18). Stage 1 AKI was defined as a 1.5- to twofold increase or 0.3-mg/dl increase in serum creatinine from day 0 to peak creatinine value on either day 1 or 2.

Avoiding Bias

We used rigorous inclusion criteria and a prospective study design to help avoid selection bias, as the outcome of AKI was unknown at time of enrollment. Furthermore, to avoid exposure and outcome misclassification, we used objective measures as our exposures, and we defined the outcome of AKI using well established criteria. We also selected runners who were not exposed to NSAIDs perimathon and did not have prior comorbidities that would increase their risk for having the outcome of AKI.

Statistical Analyses

We compared the mean (SD) or median (interquartile range) for continuous variables and frequency (percentage) for categorical variables. For variables collected using the Zephyr device, we created a plot against time to portray the trend in core body temperature throughout the race. We stratified the cohort by AKI status to compare differences in vital signs, clinical laboratory measurements, and novel biomarkers of kidney injury measured on day 1. We

compared fold change in biomarkers on day 1 compared with day 0. Lastly, we assessed the linear dependence between copeptin, sweat sodium and volume, and core body temperature with selected clinical laboratory measurements using the Spearman Correlation Coefficient. Statistical significance was defined as *P* value <0.05 using the Wilcoxon rank sum and chi-squared tests to compare AKI and non-AKI groups and the Kruskal–Wallis test to compare time points. Because of our small sample size, we evaluated unadjusted descriptive analyses and were not able to adjust for covariates or assess associative relationships between our exposure variables and the outcome of AKI. We used SAS 9.4 software (SAS Institute, Cary, NC) for all statistical analyses.

Results

Baseline Characteristics and Demographics of Cohort

A total of 23 runners were enrolled in our study (Supplemental Figure 1). Runners had a median age of 37 years old, and 43% were men (Table 1). Runners had no comorbidities, such as hypertension, cardiovascular disease, convulsive disorders, kidney disease, nephrolithiasis, or gastrointestinal disorders; ran three marathons in their lifetime on average; and finished the race in a median of 4.24 (3.59–5.05) hours. Twelve (55%) runners developed AKI, with the majority of runners developing stage 1 AKI, and only two had stage 2 AKI (Supplemental Figure 2). Seventy-four percent had a positive urine microscopy for acute tubular injury. Runners with AKI had a median of 10 (6–20) years of running experience compared with 6 (5–9) years in runners without AKI (*P*=0.04). Overall vital signs and clinical laboratory measurements shown in Table 2 showed no significant differences between runners with and without AKI, except for lower median systolic BP in those with AKI (103 [93–107] mm Hg) compared with those without AKI (114 [111–123] mm Hg; *P*=0.006) immediately after the race.

Primary Variables

Copeptin. Copeptin significantly increased on day 1 as shown in Supplemental Figure 3 20-fold (three- to 25-fold) from baseline. Furthermore, copeptin concentrations immediately after the race were significantly higher in runners with AKI compared with those without AKI (Figure 2, Table 2), having a 24-fold (eight- to 34-fold) increase in runners with AKI compared with a fourfold (three- to 17-fold) increase in those without AKI. Copeptin on day 1 did not correlate with any of the other primary variables, including sweat sodium, sweat volume, and core body temperature (Table 3). Copeptin was correlated with serum osmolality (*r*=0.56, *P*=0.008) and urine osmolality (*r*=0.41, *P*=0.07) on day 1.

Sweat Sodium and Volume. Sweat sodium loss was highly variable among runners, ranging from 0.50 to 7.20 g throughout the race. Estimated median sweat volume loss was 2.47 (1.02–3.89) L throughout the race, with a peak of 6.81 L. Comparing sweat sodium loss with sodium intake reported by runners, 20 (87%) runners had a median sodium deficit of 2.0 (0.78–3.42) g, and two (9%) runners had positive sodium balances of 0.24 and 0.46 g as shown in Supplemental Figure 4. Furthermore, when comparing sweat volume loss with fluid intake, 16 (70%) runners had a median fluid deficit of 0.99 (0.52–3.04) L, and seven (30%) runners had a positive median fluid balance of 0.94 (0.65–1.14) L.

Table 1. Demographics and race variables of 23 runners in the 2017 Hartford Marathon by AKI status

Variable	Total, n=23	No AKI, n=10	AKI, n=12
Demographics			
Age, yr	37 (35–44)	37 (32–42)	42 (35–46)
Men	10 (43%)	6 (60%)	4 (33%)
BMI, kg/m ²	24 (22–25)	25 (22–25)	22 (21–25)
Weight, kg	68 (61–77)	73 (61–80)	64 (59–68)
Height, cm	173 (163–178)	175 (165–178)	168 (163–178)
NSAIDs>48 h before race	10 (43%)	5 (50%)	5 (42%)
NSAIDs>24 h postrace	7 (30%)	5 (50%)	2 (17%)
Herbal supplementation	12 (52%)	5 (50%)	7 (58%)
Training and race variables			
Finishing pace, min/km	6.26 (5.81–7.30)	6.71 (6.11–7.30)	5.81 (5.21–7.15)
Midrace pace, min/km	6.11 (5.51–6.86)	6.26 (5.66–7.00)	5.51 (5.22–6.56)
Fastest marathon time, h	4.07 (3.75–4.75)	4.58 (4.07–4.65)	3.82 (3.45–4.77)
No. of lifetime marathons	3 (1–9)	1 (0–6)	6 (2–21)
Running experience, yr	9 (5,12)	6 (5–9)	10 (6–20)
Average weekly miles	29 (17–36)	29 (17–32)	28 (17–39)
Marathon finishing time, h	4.24 (3.59–5.05)	4.39 (4.15–5.05)	3.83 (3.40–5.00)
19+ km runs per mo	3.5 (2.5–5.0)	3 (2.5–4)	4 (3–7)

Data are shown as median (interquartile range), and frequencies are presented as *n* (%). The median age of runners was 37 years old, and 43% were men. There were no significant differences in baseline characteristics, such as weight and body mass index (BMI), between runners who developed AKI and those who did not. In contrast, some training parameters were significantly different between the two groups. Runners with more years of running experience developed AKI compared with those with less experience. There was a trend of higher participation in marathons as well as higher long-mile runs (>12 miles) in runners who developed AKI. This suggests that runners with AKI tend to be more experienced and have more extensive training. NSAID, nonsteroidal anti-inflammatory drug.

Sweat volume losses were significantly higher in runners with AKI ($P=0.03$), and sweat sodium losses had a significant trend between runners with and without AKI ($P=0.06$) as shown in Figure 2 and Table 2. Sweat sodium and volume losses significantly correlated ($r=0.88$, $P<0.001$) and sweat sodium loss as well as sweat volume loss positively correlated with net weight loss ($r=0.43$, $P=0.05$ and $r=0.56$, $P=0.006$, respectively) as shown in Table 3.

Core Body Temperature. Temperatures ranged from 35.8 to 41°C during the race, with the highest temperatures occurring in the second quartile of the race as shown in Supplemental Figure 5. There was no significant difference in temperature between runners with and without AKI as shown in Table 2. Core body temperature did not correlate with any of the other primary variables as shown in Table 3.

Secondary Variables: Novel Biomarkers of Kidney Injury

Plasma and urine biomarkers significantly increased from baseline to day 1 (Supplemental Table 3). Only plasma IL-12 was marginally higher in runners with AKI at baseline as shown in Supplemental Table 4. Plasma NGAL, YKL-40, and IL-2 levels measured immediately after race were also significantly different between runners with AKI and those without as seen in Table 4. As for urine biomarkers, both MCP-1 and YKL-40 were significantly higher in runners with AKI compared with those without AKI when adjusted for urine creatinine. Plasma IL-18 and both plasma and urine NGAL had a significantly higher fold change in runners with AKI compared with those without AKI (Supplemental Table 5).

Discussion

In this prospective study, we interrogated the role of volume and thermoregulatory responses in runners with

and without AKI to further advance our prior findings that transient ischemic AKI is common after running (4). We found that copeptin, sweat volume, and sweat sodium were significantly different between runners with and without AKI but that core body temperature was not significantly different. Our study highlights that there are hemodynamic differences in runners with and without AKI and provides insights for future research to assess if adequate sodium and fluid replenishment during the race may optimize hemodynamics and reduce the incidence of runners' AKI.

The release of copeptin in marathon runners is likely multifactorial, secondary to both osmolar and nonosmolar stimuli (19,20). In our study, we were not able to capture any significant differences in serum osmolality between runners with and without AKI. However, systolic BP was lower among runners with AKI and could have contributed to copeptin production secondary to decrease in blood flow. Additionally, given the net negative fluid balance that most runners experienced during the race as well as the substantial shunting of blood away from main organs, such as the heart and the kidneys, to enhance blood flow to the musculoskeletal system (7), it is likely that dehydration and reduction in blood flow were the driving components of copeptin release. Elevated concentrations of copeptin are prognostic markers in many clinical conditions, such as septic shock, pneumonia, cardiovascular disease, and diabetes (21–23). Furthermore, copeptin is elevated in CKD and proteinuria (24,25). Although this association could be attributed to decreased filtration by the kidneys, it has been shown that kidney clearance is not a main determinant of copeptin concentrations, because levels were unchanged in living kidney donors despite reduction in eGFR (26). In our study, it is difficult to identify if increases in copeptin

Table 2. Distribution of vital signs and clinical laboratory measurements of 23 runners in the 2017 Hartford Marathon by AKI status

Variable	Total, n=23	No AKI, n=10	AKI, n=12	P Value
1 d before race (day 0)				
Heart rate	64 (57–73)	69 (60–76)	63 (55–69)	0.16
Systolic BP, mm Hg	132 (121–137)	130 (121–132)	134 (121–138)	0.51
Serum creatinine, mg/dl	0.90 (0.75–0.98)	0.87 (0.76–0.98)	0.88 (0.75–0.96)	0.87
BUN-to-creatinine ratio	17 (15–20)	19 (15–22)	17 (13–19)	0.31
Serum sodium, mmol/L	139 (138–140)	140 (139–141)	139 (138–140)	0.08
CPK, U/L	105 (68–147)	109 (64–167)	91 (73–144)	0.82
Hemoglobin, g/dl	15 (14–16)	15 (14–16)	15 (13–16)	0.65
Copeptin, pmol/L	3.12 (2.69–3.44)	3.12 (2.69–3.36)	3.12 (2.71–3.92)	0.86
Soon after race (day 1)				
Heart rate	107 (101–126)	112 (104–127)	107 (100–122)	0.70
Systolic BP, mm Hg	107 (102–115)	114 (111–123)	103 (93–107)	0.006
Serum creatinine, mg/dl	1.28 (1.06–1.44)	1.03 (0.96–1.19)	1.42 (1.28–1.59)	0.001
BUN-to-creatinine ratio	15 (13–18)	17 (13–23)	15 (12–18)	0.21
Serum sodium, mmol/L	141 (139–143)	141 (140–143)	141 (139–143)	0.81
CPK, U/L	286 (239–403)	286 (250–423)	281 (211–342)	0.59
Hemoglobin, g/dl	14 (14–16)	14 (14–15)	15 (14–16)	0.76
Urine microscopy score >1	17 (74%)	9 (90%)	8 (67%)	0.31
Urine protein, mg/dl	15 (65%)	7 (70%)	8 (67%)	0.89
Serum osmolality, mosm/kg	303 (296–310)	303 (296–308)	303 (298–310)	0.74
Urine osmolality, mosm/kg	523 (348–651)	500 (156–714)	564 (505–573)	0.74
Copeptin, pmol/L	49.8 (11.3–81.5)	11.3 (6.6–43.7)	79.9 (25.2–104.4)	0.02
Sweat sodium loss, g	2.34 (1.15–4.2)	1.4 (0.97–2.8)	3.41 (1.7–4.8)	0.06
Sweat volume loss, L	2.47 (1.02–3.89)	1.66 (0.72–2.84)	3.89 (1.49–5.09)	0.03
Median core body temperature, °F	38.4 (38.4–39.3)	38.8 (38.4–39.4)	38.4 (38.4–38.6)	0.62
Peak core body temperature, °F	39.2 (38.6–40.2)	39.0 (38.6–40.3)	39.4 (39.0–40.2)	0.55
Total fluid intake, L	1.5 (1.2–2.0)	1.7 (1.4–2.5)	1.8 (0.9–2.0)	0.41
Total sodium intake, g	0.57 (0.26–0.96)	0.57 (0.26–0.96)	0.57 (0.22–0.91)	0.96
Net weight loss, kg	1.8 (1.5–2.5)	1.9 (1.5–2.3)	1.6 (1.1–2.9)	0.76
Net fluid balance, L	−0.67 (−2.95–0.65)	−0.06 (−3.62–0.94)	−0.96 (−2.22 to −0.28)	0.41
Net sodium balance, g	−1.89 (−3.39 to −0.66)	−1.16 (−3.34 to −0.25)	−2.22 (−3.43 to −1.81)	0.19
1 d after race (day 2)				
Heart rate	62 (57–72)	67 (59–73)	60 (57–69)	0.34
Systolic BP, mm Hg	128 (121–142)	129 (116–149)	125 (122–140)	0.45
Serum creatinine, mg/dl	0.89 (0.75–0.93)	0.88 (0.80–0.93)	0.89 (0.73–0.93)	0.64
BUN-to-creatinine ratio	20 (15–23)	17 (15–22)	21 (18–23)	0.19
Serum sodium, mmol/L	137 (136–138)	138 (136–138)	137 (135–138)	0.22
CPK, U/L	856 (542–1621)	788 (474–1455)	1339 (616–2063)	0.22
Hemoglobin, g/dl	14 (14–15)	14 (13–15)	14 (14–16)	0.52
Copeptin, pmol/L	3.57 (2.73–7.03)	3.56 (2.51–4.01)	4.78 (3.35–7.05)	0.37

In healthy adults, 13 pmol/L represents the 97.5th percentile of copeptin. Data are shown as median (interquartile range), and frequencies are presented as *n* (%). *P* values were calculated using the Wilcoxon rank sum test. Runners with AKI had significantly lower systolic BPs immediately after the race with higher copeptin levels as well as higher sweat volume losses and a trend toward higher sweat sodium losses. Core body temperature was not significantly different between runners with and without AKI. This suggests that runners with AKI had hypovolemia as evidenced by increased sweat losses and lower systolic BP leading to copeptin stimulation and likely, hypoperfusion to the kidneys secondary to kidney vasoconstriction in the setting of hypovolemia and renin-angiotensin-aldosterone system activation. CPK, creatine phosphokinase.

preceded creatinine rise, because both measurements were done simultaneously, but there are several speculated mechanisms that could explain the potential adverse effects of copeptin on kidney function. As copeptin increases, there is an increase in urine osmolality and reduction in urinary flow, which could lead to granular cast accumulation and an inflammatory kidney state (27,28). Vasopressin, for which copeptin is a surrogate marker, also activates the renin-angiotensin-aldosterone system, which leads to significant kidney vasoconstriction and further sustained reduction in blood flow to the kidneys, potentially leading to ischemic damage (29,30). Finally, administration of vasopressin has shown to exert a dose-dependent increase in kidney oxygen consumption, which may predispose to kidney hypoxia (31).

Similar to copeptin, core body temperature, sweat volume, and sweat sodium showed substantial deviations from normal resting physiology. Sweat sodium and volume losses were substantially higher in runners with AKI, but in contrast, core body temperature did not differentiate between runners with and without AKI. However, our sample size was small, and modest associations may still exist. Nonetheless, given the mild ambient temperatures during the race in Hartford, Connecticut (with an average ambient temperature of 18°C and average humidity of 84%), it is possible that core body temperature did not rise to critical levels that would play a role in AKI. Therefore, an association may still exist in hotter ambient temperatures.

We have also shown that novel biomarkers of injury and repair increase secondary to marathon running, validating

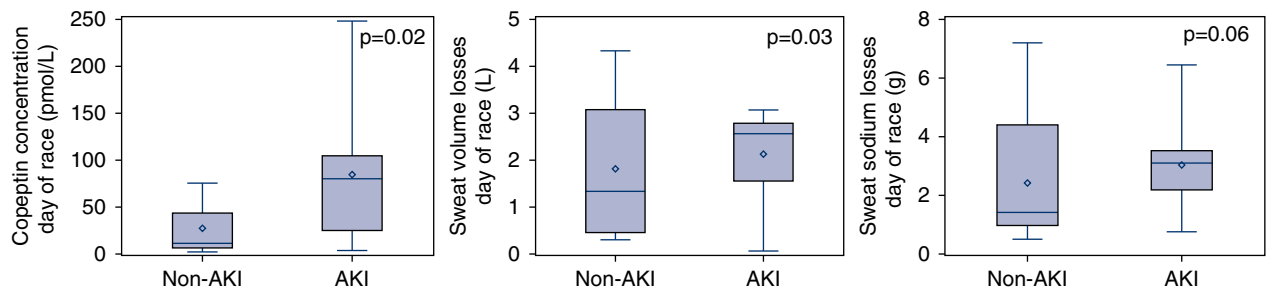


Figure 2. | Copeptin, sweat sodium, and volume losses differentiated between runners with and without AKI. Copeptin levels and sweat volume were significantly higher in runners with AKI compared with those without AKI. Additionally, there was a trend toward higher sweat sodium levels in runners with AKI.

our prior findings (4) and confirming the increased inflammatory state during marathon running (32). Some novel biomarkers on day of race were significantly higher in runners with AKI compared with runners without AKI, suggesting that the sustained physical stress imposed by marathon running leads to an acute inflammatory response that can distinguish between runners with and without AKI.

Our study is the first to investigate the possible mechanisms of AKI in marathon runners by testing markers of both volume and thermoregulatory processes. Furthermore, we are first to measure copeptin concentrations in the setting of marathon running and AKI. We have previously shown that AKI in runners is secondary to ischemic injury given findings of granular casts on microscopy as well as increase in biomarkers of tubular injury. In this study, we have identified that the cause of this ischemic injury is likely hypoperfusion secondary to hypovolemic changes during the race as evidenced by higher sweat losses and copeptin concentrations in runners with AKI. This study provides a layer of knowledge to aid in the understanding of runner AKI for future studies.

Additionally, most runners had a net negative sodium and fluid balance, highlighting the lack of adequate hydration and salt intake throughout the race. This raises an important clinical question regarding the role of volume regulatory parameters to guide personalized rehydration as

calculated by sweat losses and copeptin release to possibly prevent runners' AKI. Other studies have also shown that runners are inadequately hydrating during the race, and >70% of runners feel that dehydration plays a role in their race performance (33,34). Guidelines regarding hydration during a marathon are limited, because most studies rely on subjective data (33), and there are no randomized, controlled trials to help guide hydration protocols (35). Our study provides a more detailed understanding of the hemodynamic state of runners throughout the race by the measurement of objective volume-regulatory markers. These markers may be used in future research to more accurately guide adequate hydration protocols in runners and other endurance athletes. Our study also identified that runners with AKI had more years of running experience and more participation in marathons as well as more extensive training. Although a prospective longitudinal study of runners is needed, these data are suggestive that, with more marathon exposure, there may be a higher risk of AKI and that experienced runners are a potential target group for future studies. However, it is important to note that the assessment of kidney injury in runners requires a multifaceted approach involving many factors, such as change in kidney injury biomarkers and urine microscopy along with a rise in serum creatinine, to determine true injury.

Table 3. Correlations between primary exposure variables, systolic BP, net weight loss, and Δ creatinine on the day of the race

Core Body Temperature, °F	Core Body Temperature, °F	Sweat Sodium Loss, g	Sweat Volume Loss, L	Copeptin, pmol/L	Systolic BP, mm Hg	Net Weight Loss, kg	Δ Creatinine, mg/dl	Serum Osmolality
Sweat sodium loss, g	-0.06 ($P=0.81$)							
Sweat volume loss, L	-0.13 ($P=0.56$)	0.88 ($P<0.001$)						
Copeptin, pmol/L	-0.12 ($P=0.60$)	0.17 ($P=0.46$)	0.13 ($P=0.58$)					
Systolic BP, mm Hg	-0.12 ($P=0.60$)	-0.41 ($P=0.06$)	-0.09 ($P=0.68$)	-0.35 ($P=0.12$)				
Net weight loss, kg	-0.06 ($P=0.79$)	0.43 ($P=0.05$)	0.56 ($P=0.006$)	0.005 ($P=0.98$)	-0.17 ($P=0.44$)			
Δ Creatinine, mg/dl	-0.05 ($P=0.82$)	0.14 ($P=0.54$)	-0.03 ($P=0.90$)	0.21 ($P=0.36$)	-0.37 ($P=0.09$)	-0.16 ($P=0.48$)		
Serum osmolality	0.003 ($P=0.99$)	0.24 ($P=0.29$)	0.17 ($P=0.46$)	0.56 ($P=0.008$)	-0.36 ($P=0.10$)	0.34 ($P=0.12$)	0.25 ($P=0.26$)	
Urine osmolality	-0.23 ($P=0.29$)	-0.24 ($P=0.28$)	-0.22 ($P=0.31$)	0.41 ($P=0.07$)	-0.02 ($P=0.92$)	0.21 ($P=0.34$)	0.18 ($P=0.43$)	0.27 ($P=0.23$)

Table 4. Nontraditional plasma and urine biomarkers of kidney injury of 23 runners in the 2017 Hartford Marathon by AKI status

Biomarker	Total, n=22	No AKI, n=10	AKI, n=12	P Value
Plasma biomarkers day of race				
IFN, pg/ml	2.43 (1.90–3.61)	2.19 (1.89–3.61)	3.1 (1.99–3.97)	0.39
IL-2, pg/ml	0.22 (0.14–0.29)	0.14 (0.11–0.21)	0.28 (0.16–0.50)	0.04
IL-4, pg/ml ^a	0.03 (0.02–0.04)	0.025 (0.02–0.04)	0.03 (0.01–0.04)	0.67
IL-6, pg/ml	21.96 (9.98–33.73)	18.67 (8.98–30.74)	25.71 (11.75–35.59)	0.29
IL-8, pg/ml	15.3 (11.9–21.14)	12.18 (9.57–14.09)	19.36 (13.12–23.47)	0.05
IL-10, pg/ml	11.04 (1.38–16.81)	5.33 (0.98–11.98)	12.37 (4.47–21.13)	0.15
IL-12, pg/ml ^a	0.22 (0.14–0.30)	0.20 (0.14–0.30)	0.23 (0.14–0.28)	>0.99
IL-13, pg/ml ^a	0.62 (0.46–1.13)	0.60 (0.46–1.13)	0.65 (0.34–1.01)	0.97
IL-18, pg/ml	280 (247–338)	270 (246–320)	307 (255–461)	0.29
KIM-1, pg/ml	88.13 (73.63–139.02)	79.64 (64.61–90.96)	124.80 (85.3–167.71)	0.11
NGAL, ng/ml	143.54 (122.27–200.78)	122.40 (107.59–140.08)	189.90 (142.18–292.25)	0.01
TNF- α , pg/ml	2.84 (2.22–3.05)	2.32 (1.97–3.05)	2.92 (2.61–4.53)	0.07
MCP-1, pg/ml	378.02 (312.45–486.45)	346 (270.26–452.33)	410.69 (325.20–554.12)	0.15
YKL-40, ng/ml	36.98 (32.54–50.61)	32.79 (30.43–35.93)	50.61 (35.92–64.83)	0.02
Urine biomarkers day of race (raw concentrations)				
IL-18, pg/ml	74.72 (40.87–180.56)	49.28 (40.87–113.20)	104.83 (23.25–209.03)	0.35
KIM-1, pg/ml	2680 (741–3314)	1841 (519–2769)	2890 (1165–4862)	0.21
NGAL, ng/ml	44.14 (15.99–100.53)	25.94 (16.60–77.78)	44.14 (15.81–128.74)	0.75
MCP-1, pg/ml	864 (143–1635)	314 (143–864)	1210 (138–2557)	0.13
YKL-40, pg/ml	2114 (611–4008)	1459 (565–2114)	3534 (1484–6593)	0.07
Urine biomarkers day of race (indexed to urine creatinine)				
IL-18/creatinine, pg/mg	38.51 (29.14–69.92)	33.27 (29.14–45.64)	38.52 (28.73–86.48)	0.58
KIM-1/creatinine, pg/mg	1208 (741–1459)	791 (631–1315)	1256 (944–1804)	0.08
NGAL/creatinine, ng/mg	33.59 (13.85–47.06)	32.38 (13.28–39.88)	30.05 (14.22–81.40)	0.62
MCP-1/creatinine, pg/mg	302 (186–747)	187 (175–219)	467 (281–923)	0.02
YKL-40/creatinine, pg/mg	941 (674–1853)	762 (537–941)	1612 (926–2698)	<0.006

Data are shown as median (interquartile range). P values were calculated using the Wilcoxon rank sum test. Injury biomarkers, such as IL-2 and neutrophil gelatinase-associated lipocalin (NGAL), were significantly elevated in runners with AKI compared with runners without AKI, and the repair biomarker chitinase-3-like 1 (YKL-40) was also significantly elevated in those with AKI. This may suggest that runners with AKI have higher levels of physical stress during the race, leading to systemic release of injury and repair biomarkers. KIM-1, kidney injury molecule-1; MCP-1, monocyte chemoattractant protein-1.

^aValues below the limit of detection (BLD) were excluded from analysis. IL-12 has two and one value BLD in the AKI and non-AKI groups, respectively. IL-13 has one value BLD in both AKI and non-AKI groups. IL-4 has one and four values BLD in the AKI and non-AKI groups, respectively.

We also acknowledge the limitations of our study, which include small sample size, and hence, our results are subject to confounding. We did not capture any long-term outcomes due to our short follow-up time of 3 days. Additionally, our total sweat sodium and volume measurements were calculated from the 5-mile mark measurements, which presume constant sweat and fluid losses throughout the race and may not be accurate. It is likely that sweat and fluid losses increase with increasing intensity and pace of running, and hence, our calculations may have underestimated the actual sweat and fluid losses throughout the race, which would bias our results toward the null. Our net fluid and sodium balance values were calculated *via* information from a postsurvey questionnaire, and the accuracy of the data may be subject to recall bias. We were also unable to quantify the urine output of runners in our study. The generalizability of our study is limited to moderate ambient temperatures and may not apply to races performed in hotter temperatures. Furthermore, our cohort consisted of experienced runners, and our findings may not apply to runners with limited running experience. However, our study did find that, among experienced runners, there was a higher incidence of AKI in those with longer history of running. It is possible that experienced runners had more

years of training as well as more exposure to marathons with faster pace of running compared with less experienced runners. There could be cumulative subclinical recurring kidney injury in experienced runners, leading to less kidney reserve and higher risk of reinsult with additional marathons. The metabolic demands are heightened during a marathon, and with a faster pace of running, there is likely a further increase in metabolic demands and hemodynamic shifts, leading to higher risk of AKI. Furthermore, many studies have shown that marathon runners and endurance athletes have cardiac fibrosis (36–39), and it is plausible that kidney fibrosis also occurs with frequent marathon running. Future research needs to focus on the long-term kidney effects of marathon running to better determine how marathon participation affects long-term kidney function, which our study did not address. Lastly, the strength of inference testing in our study may be affected by lack of adjustment for multiple comparisons in the context of multiple planned analyses.

To conclude, our study shows that core body temperature, sweat volume, sodium losses, and copeptin concentrations are significantly affected during marathon running but that only sweat sodium, volume losses, and copeptin concentrations differed between runners with and without

AKI. We have identified hemodynamic differences as measured by these regulatory markers between runners with and without AKI, which provide the building blocks for future research to assess if adequate hydration and optimization of hemodynamics will reduce the incidence of AKI in runners. Whether these markers play a causative role in AKI or are simply markers of injury in marathon runners with AKI warrants additional investigation.

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Both Dr. Mansour and Dr. Parikh had a role in study design, collection, analysis, and interpretation of data; writing of the report; and the decision to submit this manuscript for publication.

Disclosures

Dr. Bjornstad reports consultancy fees from Bayer, Bristol-Myers Squibb, and Boehringer Ingelheim and declares an unpaid position on the advisory board of XORTX. Dr. Parikh reports consulting fees from Akebia Therapeutics, Inc. and Genfit Biopharmaceutical Company and other fees from RenalytixAI. Dr. El-Khoury, Dr. Jia, Dr. Kukova, Dr. Mansour, Dr. Martin, Dr. Myrick, Dr. Obeid, and Dr. Pata have nothing to disclose.

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Supplemental Material

This article contains the following supplemental material online at <http://cjasn.asnjournals.org/lookup/suppl/doi:10.2215/CJN.01400219/-/DCSupplemental>.

Supplemental Figure 1. Enrollment chart of runners in the study cohort.

Supplemental Figure 2. Serum creatinine levels per runner.

Supplemental Figure 3. Copeptin concentrations by time point.

Supplemental Figure 4. Net sodium and fluid balance during the race per runner stratified by AKI status.

Supplemental Figure 5. Series plot of core body temperature every second during the race per runner stratified by AKI status.

Supplemental Table 1. Equations used to calculate total sweat volume and sodium.

Supplemental Table 2. Coefficients of variance (CVs) for copeptin and kidney injury biomarkers.

Supplemental Table 3. Day of race biomarker levels by time point.

Supplemental Table 4. Prerace biomarker levels stratified by AKI status.

Supplemental Table 5. Fold change in biomarkers by AKI status ($n=22$, excluded runner 42).

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